

Installation Manual

Generator Set

HGLAA Spec A

English - Original Instructions



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Chapter 1 - Introduction

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1.1 About the Service Information - Owner - Overview

General Information

Cummins Inc. manuals are created to support Cummins® products.

The content of this manual is based on the information in effect at the time of publication.

- Cummins Inc. reserves the right to make any changes at any time without obligation.
- If differences are found between your product and the information in this manual, contact a Cummins® Authorized Repair Location.

For information on components or fault codes not supplied by Cummins Inc., contact the original equipment manufacturer (OEM) or supplier.

This set of literature is intended to provide guidance and recommendations for the safe operation and maintenance of the applicable Cummins® product.

This product has been designed to provide safe and efficient service when properly installed, maintained, and operated.

Overview

Owner/Operator information contains the following.

- Operating instructions
- Daily maintenance schedules
- Routine maintenance schedules
- Overhaul schedules, when applicable
- Common repair instructions
- Basic troubleshooting

1.2 Illustrations - Overview

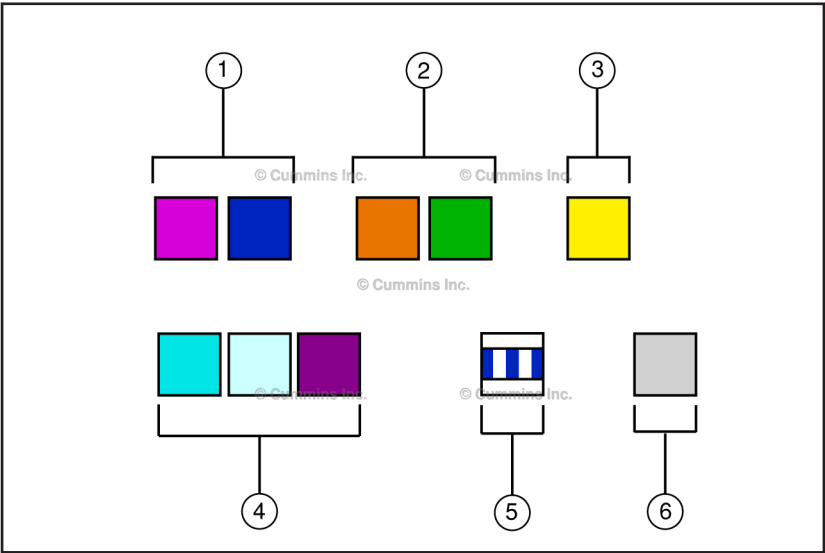
General Information

Various imagery styles are utilized throughout the service information.

- Online images are delivered in color, rendered styles.
- Printed images are delivered in black and white line art styles.

When viewing images online within instructional context, the colors represent the following actions.

- Blue or Magenta - Target, or primary component being handled.
- Orange or Green - Additional components being removed or installed to access the target component.
- Yellow - Areas highlighted for inspection, special attention, or adjustment.
- Cyan, Light Blue, or Purple - Special tools or other equipment.
- Blue/White Dash - Indicating surfaces to apply a chemical or material.
- Light Grey - Remaining, background components.



Chapter 2 - Safety

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2.1 General Safety - Generator Set RV - Overview

General Information

This procedure contains general safety precautions that must be followed to provide personal safety.

Read and understand the safety information and precautions before operating or performing any repair.

Overview

Work Environment

Follow these recommended practices when servicing products.

- Always follow on-site safety requirements and regulations.
- Always follow local training, certification, authorization, and specific customer requirements.
 - Do not work on products unless proper training has been completed to allow safe repair completion.
 - Do not operate equipment before product familiarization using the equipment operator manual.
- Always be aware of hazardous conditions that may exist in the work environment.
 - Work in a well-ventilated area away from ignition sources.
 - If adverse weather conditions are present, take appropriate safety precautions when performing work.
- Training
 - Confirm if you have received task and safety training for the job being performed.
- Working Alone
 - Avoid working alone.
 - Avoid working where you are not able to be seen or heard by another person.
 - If you must work alone, notify others of your location and schedule check-in times.
- Safety Barricades and Warnings
 - Mark overhead work areas with barricade tape or signs.
 - Mark open floor hazards with barricade tape, signs, or cones.
- Confined Space
 - Confirm if a confined space entry permit is required.
 - If required, confirm the permit is posted, signed, and dated correctly.
- Hot Work
 - Confirm a functional fire extinguisher is readily available.
 - Fire extinguishing rating must meet required safety regulations based on type of work being performed.
 - Maintain separation between ignition sources and fuel sources.
- Wheel Chocks
 - Place wheel chocks at either the front or back tire of the unit prior to starting the task.
- Spotter
 - Use a spotter when moving a customer's unit.
 - Confirm the driver can see and hear the spotter when moving.

- Housekeeping (The 5 S's - Scrap or Segregate, Set to Order, Spotless, Standardize, and Sustain)
 - Remove parts, extension cords, air hoses, and liquids from the work area that may cause trip, slip, or fall hazards.

Perform the following prior to beginning work on any products.

- Shutdown the equipment unless otherwise directed by troubleshooting procedures.
- Always allow the product to cool.
- Lock-Out-Tag-Out Equipment.
- Always make sure the product is properly supported by blocks or stands.
 - Do not work on a product supported only by lifting jacks or hoists.
- Become familiar with the tools required for performing the task at hand and how to use those tools correctly.
- Use only genuine Cummins or Cummins Recon replacement parts as instructed.

Best Safety Practices

Follow these recommended practices when servicing or operating equipment.

- Always wear protective glasses and protective shoes.
- Remove rings, watches, long jewelry, or metallic items.
- Do not wear loose fitting or torn clothing, jewelry, long hair, etc.
- Do not perform any repairs, or operate equipment, when fatigued or impaired due to drugs or alcohol.
- Always use tools that are in good condition.
- Do not work on equipment with the batteries connected or that is running unless otherwise directed by troubleshooting procedures.
- If any work must be performed while the batteries are connected or the unit is running, use extreme caution especially around hot components, moving parts, etc.
- Exercise caution when working on products that have just been turned off.
 - Hot parts may cause burns or ignite or melt common materials.
- Do not bleed the fuel system of a hot engine.
 - Contact with hot manifolds or other components may cause a fire.
- Only use proper engine barring techniques.
- Do not lift components that weigh 23 kg [50 lb] or more.
 - Use mechanical help or seek assistance.
- Exercise caution when working around rotating parts.
 - Rotating parts can cause cuts, mutilation, or strangulation.
- Exercise caution when working on electrical components.
 - High voltages can cause serious injury or death.
- Relieve system pressure as instructed before removing or disconnecting lines, fittings, or related items.
- Always test for pressure leaks as instructed.
- Always torque fittings and connections to the required specifications.
 - Over or under tightening can damage threads and create leaks.
- Always use the same fastener part number, or equivalent, when replacing fasteners.

Standard Job Safety Work Practices

- Job Safety Analysis.
 - Assess the job to identify safety hazards that may occur during the repair event.

- Ascending or Descending
 - Maintain 3 points of contact when using steps, ladders, or entering and exiting a unit.
- Communication
 - When working with others, make sure you understand what each other is doing to safely complete the task.
- Eyes On Hands and Work.
 - Confirm if you will be able to maintain an unobstructed view of your hands at all times while performing the task.
- Eyes On Path
 - Watch for hazards in your path to avoid trip or slip hazards. Examples are pits, platform edges, etc.
- Line Of Fire
 - Position yourself so that you avoid striking against, or being struck by, anything that can swing, fall, or roll.
- Pinch Point
 - Prevent exposure of all parts of your body to a nip hazard or pinch point.
- Rushing
 - Take adequate time to safely perform the job. Do not rush or take short cuts.
- Follow Procedures
 - Utilize QuickServe® Online or other standard procedures when available.
 - Make sure the procedures are correct and safe.

Job Safety Assessment

Completing a Job Safety Assessment prior to performing work helps identify job safety hazards and prevent incidents.

Use the guidelines below to assess if a situation is safe or at risk prior to performing designated work.

- If determined to be at risk, take appropriate precautions to prepare for, or eliminate, the hazard.
- If the risks are uncontrollable, consult a knowledgeable resource to find a safe practice solution.
 - A knowledgeable resource may include, but is not limited to, one of the following:
 - Site supervisor
 - Customer
 - Work supervisor
- Always check with the site where work is being performed to determine if safety assessment documentation is required.

Emergency Action Plan

Personnel engaged in installation, commissioning, operating and maintenance of Cummins Inc. products must also comply with all statutory requirements and regulations regarding health and safety.

- Establish and have in place an emergency action plan that complies to all local safety regulations.
 - Fire extinguishing media that meets required safety regulations based on type of work being performed.
 - First aid kits available.
 - Eye flushing safety device.
 - Chemical spill kit for prompt response and clean-up of a spill, discharge, or leak of hazardous materials or wastes.

Common Hazards

Follow these recommended practices when interacting with equipment as the following hazards may exist.

- High Temperature Area
 - Be alert for high temperature areas which may cause severe burns.
 - High temperature areas may be encountered in the following situations.
 - On products that have just been turned off.
 - On or around exhaust related components (turbocharger, aftertreatment systems, etc).
 - In exhaust gas flow paths.
 - Contacting hot fluid lines, tubes, or compartments.
- Recommended Practices
 - Allow components to cool before servicing.
 - Verify the temperature of the component.
 - Utilize an infrared gun, temperature sensor, temperature gauge, or other reliable method to determine component temperature.
 - Protect Eyes.
 - Protect Skin.
 - Wear insulated gloves.
 - Make sure surrounding items do not come in contact with hot components or exhaust.
 - Contact may ignite or melt those materials.
- Heavy Objects
 - Be alert when working with heavy objects.
 - Do not lift components that weigh 23 kg [50 lb] or more.
 - Use mechanical help or seek assistance.
 - Use mechanical help to move items when ever possible.
 - Make sure the load is securely fastened to the equipment.
 - Make sure lifting devices, like chains, hooks, slings, etc., are in good condition and are rated for the correct capacity before use.
 - Make sure lifting devices are positioned correctly before use.
 - Use a spreader bar when necessary.
 - If the item can be lifted manually, squat to lift and lower the item.
 - Do NOT bend at the waist.
 - Maintain balance when lifting items by keeping feet apart or staggered if possible.
 - If the item must be carried, make sure the path is clear when carrying the item to, and placing the item in, the desired location.
- Pressurized Areas
 - Be alert for pressurized areas. Pressurized areas may be encountered in the following situations.
 - Air, Oil, Fuel, and Cooling systems.
 - When disconnecting or removing lines, fittings, or related items.
 - When disconnecting a device from a pressurized system.
 - When removing or loosening caps on tanks or pressurized systems.
 - Injuries that may result when interacting with pressurized areas are listed below.
 - High pressure spray can penetrate the skin, serious injury or death may result.
 - Hot fluid spray can cause burns.

- Recommended Practices
 - Protect Skin.
 - Wear impervious gloves.
 - If skin penetration from high pressure spray occurs, it is a medical emergency, seek medical attention immediately.
 - Check for pressure leaks as instructed.
 - Never check for pressure leaks with your hand.
 - Allow product to cool before accessing pressurized areas.
 - Relieve system pressure as instructed.
 - Slowly loosen fill caps to relieve pressure before servicing.

Common Substances

Follow these recommended practices when interacting with the following substances, for information regarding proper handling of various substances refer to the manufacturer's safety data sheet.

- Storage
 - Do not store any of the following common substances in the generator set compartment.
 - Oily rags and other material can restrict cooling and cause fire.
 - Keep the generator set, drip pan, and compartment clean.
- Coolant
 - Coolant is also referred to as antifreeze.
 - Protect Eyes.
 - In case of contact with eyes, flush with water for a minimum of 15 minutes.
 - Seek medical attention immediately.
 - Protect Skin.
 - In case of contact with skin, wash with soap and water.
 - Remove contaminated clothing.
 - If injection occurs, it is a medical emergency, seek medical attention immediately.
 - Do not ingest.
 - If ingested, drink excess water for dilution and seek medical attention.
 - Do not pour used antifreeze into containers that have been used to store other chemicals or products, such as oil or gasoline, unless they have been thoroughly cleaned.
 - If material is spilled, avoid contact and dispersal with runoff, soil, waterways, drains, and sewers.
 - Provide adequate ventilation to the area.
 - Absorb with sand, clay, or commercial absorbent.
 - Transfer to containers and neutralize the material.
 - Flush spill area with soap and excess water.
 - Report spills effecting water source contamination to local authorities immediately.
 - Proper disposal is required.
 - Dispose of in accordance with local and environmental regulations.

- Liquid Nitrogen
 - Work in a well-ventilated area.
 - Protect Eyes.
 - In case of contact with eyes, flush with water for a minimum of 15 minutes.
 - Seek medical attention immediately.
 - Protect Skin.
 - In case of contact with skin, seek medical attention immediately.
 - Wear protective clothing and gloves that insulate.
 - Handle items with tongs or wire hooks.
 - Avoid prolonged breathing of liquid nitrogen vapors.
 - Utilize breathing assistance or oxygen if necessary.
- Lubricating Oil
 - Protect Eyes.
 - In case of contact with eyes, flush with water for a minimum of 15 minutes.
 - Protect Skin.
 - In case of contact with skin, wash with soap and water.
 - Do not ingest.
 - If ingested, seek medical attention.
 - Proper disposal is required.
 - Dispose of in accordance with local and environmental regulations.
- Refrigerant
 - Protect Eyes.
 - In case of contact with eyes, flush with water for a minimum of 15 minutes.
 - In case of frostbite, use lukewarmwater, not hot.
 - Seek medical attention if irritation continues.
 - Proper disposal is required.
 - Dispose of in accordance with local and environmental regulations.
 - Only disconnect liquid refrigerant lines in a well-ventilated area.
 - Liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere.
 - Federal law requires capturing and recycling refrigerant in the United States of America.
- Solvents
 - Follow the manufacturer's instructions for safe handling practices.
 - Follow the manufacturer's recommendations for use.
 - Some solvents are flammable and toxic.
 - Protect Eyes.
 - In case of contact with eyes, follow manufacturer's recommendations.
 - Protect Skin.
 - In case of contact with skin, follow manufacturer's recommendations.
 - Dispose of in accordance with manufacturer's recommendations.

- Starting Aids (Starting Fluid)
 - Do NOT use starting fluid if the intake air heater option is used.
 - Do NOT use volatile cold starting aids in underground mine or tunnel operations.
 - The local United States Bureau of Mines inspector can provide more information and instructions.
 - Avoid sparks, arcing switches and equipment, cigarettes, pilot lights, flames, and other sources of ignition.
 - Work in a well-ventilated area.
 - Avoid inhalation.

Electrical Components

Follow these recommended practices when interacting with electrical components.

- Lead Acid Batteries
 - Protect Eyes.
 - Wear safety glasses or goggles.
 - In case of battery acid contact with eyes, flush with water for a minimum of 15 minutes.
 - Seek medical attention immediately.
 - Protect Skin.
 - Wear rubber gloves and a chemical apron.
 - In case of battery acid contact with skin or clothing, rinse with water for several minutes.
 - Avoid spreading the acid.
 - Seek medical attention immediately.
 - Do NOT open the battery caps with your face over or near the battery.
 - Remove rings, watches, long jewelry, or metallic items when working with or near batteries.
 - Ventilate the battery compartment before servicing the battery.
 - Work in a well-ventilated area.
 - Avoid sparks, arcing switches and equipment, cigarettes, pilot lights, flames, and other sources of ignition.
 - Use insulated or non-conducting tools.
 - Neutralize static buildup by contacting the nearest ground surface before working on a battery.
 - Do not lift batteries by the posts.
 - Do not touch both battery terminals with your bare hands at the same time.
 - Disconnect the negative (-) battery cable first.
 - Attach the negative (-) battery cable last.

Fuels

Follow these recommended practices when interacting with equipment that uses different types of fuels.

- General recommended safety practices for all fuels.
 - Protect Eyes.
 - Protect Skin.
 - ALWAYS test for fuel leak as instructed.
 - Do NOT dilute.
 - Avoid sparks, arcing switches and equipment, cigarettes, pilot lights, flames, and other sources of ignition.
 - Provide extra ventilation to the work area.
 - Do not troubleshoot or repair fuel leaks while the engine is running.
 - If material is spilled, avoid contact and dispersal with runoff, soil, waterways, drains, and sewers.
 - Absorb with sand, clay, or commercial absorbent.
 - Transfer to containers and neutralize the material.
 - Flush spill area with soap and excess water.
 - Report spills effecting water source contamination to local authorities immediately.
 - Proper disposal is required.
 - Dispose of in accordance with local and environmental regulations.
 - Always torque fittings and connections to the required specifications.
 - Over or under tightening can damage threads and create leaks.
 - Storage
 - Do not store any fuels in the generator set compartment.
 - Oily rags and other material can restrict cooling and cause fire.
 - Keep the generator set, drip pan, and compartment clean.
- Gasoline additional recommended safety practices.
 - ALWAYS be alert for the smell of gasoline.
 - Vapors accumulate near the floor.
 - Check the work floor, sumps, and low-lying areas for ignition sources before servicing equipment.
- Biodiesel additional recommended safety practices.
 - Vapors accumulate near the floor.
 - Check the work floor, sumps, and low-lying areas for ignition sources before servicing equipment.

- Compressed Natural Gas additional recommended safety practices.
 - Always be alert for the smell of gas.
 - Compressed natural gas is typically treated with an odor producing chemical for leak detection.
 - Non-refined sources of natural gas (landfill gas, biogas, coal bed gas, wellhead gas, etc.) can not always be detected by smell.
 - Always test for fuel leaks as instructed. Odorant can fade.
 - Upon entering a room or approaching a vehicle where the smell of gas is present, immediately shutoff all engines and ignition sources.
 - Natural gas ignites when there is a 5% - 15% mixture in the air.
 - Asphyxiation can occur when concentration reaches 21% or more.
 - Do not start equipment or nearby equipment until a suspected gas leak is corrected and the area is ventilated.
 - Work in areas that do not share common ventilation with areas containing ignition sources.
 - Store and service natural gas fueled equipment in large, well-ventilated areas, or outside.
 - Natural gas accumulates near the ceiling.
 - Check the ceiling of the work area for ignition sources before servicing equipment.
 - Only disconnect gas lines in a well-ventilated area.
 - Natural gas ignition systems produce high voltage during operation.
 - Do not touch ignition wiring or components while the engine is operating.
 - If necessary, use only insulated tools.
 - Natural gas exhaust systems operate at higher temperatures than similar diesel exhaust systems.
 - Do not touch exhaust components.
 - Do not route lines or hoses which deteriorate from heat exposure near exhaust components or in the flow path of the exhaust.
 - Upon entering a room or approaching a vehicle where the smell of gas is present, immediately shutoff all engines and ignition sources.

- Liquefied Natural Gas additional recommended safety practices.
 - Always be alert for the smell of gas.
 - Non-refined sources of natural gas (landfill gas, biogas, coal bed gas, wellhead gas, etc.) can not always be detected by smell.
 - Liquefied natural gas may not have an odor.
 - Always test for fuel leaks as instructed. Odorant can fade.
 - Natural gas ignites when there is a 5% - 15% mixture in the air.
 - Asphyxiation can occur when concentration reaches 21% or more.
 - Do not start equipment or nearby equipment until a suspected gas leak is corrected and the area is ventilated.
 - Work in areas that do not share common ventilation with areas containing ignition sources.
 - Store and service natural gas fueled equipment in large, well-ventilated areas, or outside.
 - Natural gas accumulates near the ceiling.
 - Check the ceiling of the work area for ignition sources before servicing equipment.
 - Only disconnect gas lines in a well-ventilated area.
 - Natural gas ignition systems produce high voltage during operation.
 - Do not touch ignition wiring or components while the engine is operating.
 - If necessary, use only insulated tools.
 - Natural gas exhaust systems operate at higher temperatures than similar diesel exhaust systems.
 - Do not touch exhaust components.
 - Do not route lines or hoses which deteriorate from heat exposure near exhaust components or in the flow path of the exhaust.
 - Liquefied natural gas is stored in vehicle tanks at extremely cold temperatures.
 - If there is a liquefied natural gas spill, evacuate the area immediately and do not attempt to make contact with the liquid.
 - Vapors accumulate near the floor.
 - Check the work floor, sumps, and low-lying areas for ignition sources before servicing equipment.

- Liquefied Petroleum Gas
 - Always be alert for the smell of gas.
 - Liquefied petroleum gas is typically treated with an odor producing chemical for leak detection.
 - Always test for fuel leaks as instructed. Odorant can fade.
 - Upon entering a room or approaching a vehicle where the smell of gas is present, immediately shutoff all engines and ignition sources.
 - Work in areas that do not share common ventilation with areas containing ignition sources.
 - Store and service natural gas fueled equipment in large, well-ventilated areas, or outside.
 - Liquefied petroleum gas accumulates near the floor.
 - Check the work floor, sumps, and low-lying areas for ignition sources before servicing equipment.
 - Liquefied petroleum gas ignition systems produce high voltage during operation.
 - Do not touch ignition wiring or components while the engine is operating.
 - If necessary, use only insulated tools.
 - Liquefied petroleum gas exhaust systems operate at higher temperatures than similar diesel exhaust systems.
 - Do not touch exhaust components.
 - Do not route lines or hoses which deteriorate from heat exposure near exhaust components or in the flow path of the exhaust.
 - Liquefied natural gas is stored in vehicle tanks at extremely cold temperatures.
 - If there is a liquefied natural gas spill, evacuate the area immediately and do not attempt to make contact with the liquid.

Personal Protective Equipment (PPE)

- To reduce the possibility of personal injury, personal protective equipment (PPE) should be utilized.
 - Always meet on-site safety regulations for required PPE.
 - Proper maintenance of safety equipment must be practiced.
 - Integrity of safety equipment must be checked to make sure equipment functionality is maintained.
- Eye Protection
 - Eye protection must always be worn. Types of eye protection to consider are listed below.
 - Safety glasses. Exposure to flying particles or debris, chemicals or caustic liquids, gases or vapors.
 - Polarized safety glasses. Working in outdoor or bright lighting environments.
 - Over-the-glass safety glasses. Add protection to prescription glasses.
 - Safety goggles: Handling caustic liquids or chemicals.
 - Shade or arc rated eyewear. Exposure to welding. Use appropriate filter ratings.

- Foot Protection
 - Protective shoes must always be worn. Wear appropriate foot protection based on the task being completed. Types of protective footwear to consider are listed below.
 - Steel toed shoes. Exposure to falling or rolling objects. Working with or around parts, tools, and equipment.
 - Chemical resistant. Exposure to chemicals and other fluids.
 - Overshoes and overboots. Add protection to everyday work shoes.
 - Foot, toe, and metatarsal guards. Add protection to everyday work shoes.
 - Electrical hazard safety toe shoes. Exposure to electrical hazards.
 - Leather footwear or shoe protectors. Exposure to welding or arc flash.
 - Cold protection. Exposure to cold weather.
- Head and Face Protection
 - Wear appropriate face protection based on the task being completed. Types of head and face protection to consider are listed below.
 - Hard hats. Exposure varies. Consider welding, heat, or arc-rated.
 - Visors. Exposure varies. Consider welding, heat, or arc-rated.
 - Face liners. Exposure to cold weather.
 - Face shields: Exposure to liquid splash. Handling caustic liquids or chemicals.
- Hand Protection
 - Wear appropriate type and fit of gloves based on the task being completed. Types of protective gloves to consider are listed below.
 - Heat resistant or insulated. Exposure to hot items.
 - Flame resistant. Exposure to welding or arc flash.
 - Impact resistant. Performing repetitive impact and vibration work. Using pneumatic tools.
 - Impervious. Exposure to high pressure fluids.
 - Chemical resistant. Exposure to chemicals, fluids, or batteries.
 - Cut resistant. Handling sharp objects or tools.
 - Cold weather. Exposure to cold weather.
- Hearing Protection
 - When working around operating equipment, appropriately rated hearing protection should be worn. Types of hearing protection to consider are listed below.
 - Single use ear plugs.
 - Pre-formed ear plugs.
 - Ear muffs.
- Protective Clothing
 - Wear appropriate protective clothing based on the task being completed. Types of protective clothing to consider are listed below.
 - Flame resistant. Exposure to electrical hazards. Exposure to oil and gas or generator set applications. Performing welding.
 - Chemical resistant. Exposure to chemicals.
 - High visibility. Exposure to reduced visibility working environments. Working on mining, oil and gas, or sites with large equipment.

- Respiratory Protection
 - Wear appropriate respiratory protection based on the task being completed. Types of respiratory protection to consider are listed below.
 - Disposable respirators. Exposure to dust and particles, welding fumes, nuisance odors, nuisance level acid gas.
 - Reusable respirators. Exposure to cleaning, machining, welding, sanding, grinding, etc.
- Fall Protection
 - Utilize fall protection if a task is being completed more than 1.2 m [4 ft] above a solid surface. Types of fall protection to consider are listed below.
 - Fall harness and lanyard combinations.
 - Safety nets.
 - Guardrails.

Tools and Equipment

- Tool Selection
 - Select the correct tool or equipment to perform the task.
- Tool Condition
 - Confirm the tool or equipment is free of defects before use.
 - Always use tools that are in good condition.
- Tool Use
 - Use the tool or equipment as directed.
 - Follow the manufacturer's instructions.

Ergonomics

- Back-Bending and Twisting
 - Avoid bending forward more than 45° at your waist.
 - Avoid working with your back twisted with loads over 23 kg [50 lb].
 - Wear approved back support belt when appropriate.
- Lifting and Lowering
 - Squat to pick up parts.
 - Keep loads close to the body when lifting or carrying.
 - Use a team lift or a lifting device if the object is more than 23 kg [50 lb].
 - Wear approved back support belt when appropriate.
- Knee
 - Avoid bending your knee more than 90°.
 - Avoid kneeling for more than 4 hours per day.
 - Wear approved knee protective devices when appropriate.
- Pulling or Pushing
 - Pull with your arms.
 - Push with your legs.
 - Avoid exerting more force than necessary.
 - Avoid moving heavy load(s) too quickly.

2.2 Safety Messages - Overview

General Information

Safety messages are utilized throughout Cummins® Service Information to alert users of hazards that can be encountered when performing work as instructed on Cummins® products.

Safety messages define the following characteristics:

- Severity- seriousness of the hazard in terms of how much harm or damage it can cause
- Hazard- any potential source or agent that can cause harm or damage to life or health
- Risk- negative consequences caused by exposure or interaction with the hazard
- Avoidance- procedures taken to best avoid the risk that the hazard poses

The severity of a safety message is defined by one of the following color-coded signal words:

- Danger – Red background. Indicates a hazardous situation which will result in death or serious injury.
- Warning – Orange background. Indicates a hazardous situation which could result in death or serious injury.
- Caution – Yellow background. Indicates a hazardous situation which could result in minor or moderate injury.

2.3 Batteries - Overview

General Information

The purpose of this document is to make all repair locations aware of safe lead acid battery service instructions.

There are many types of batteries with varying technology.

Common battery types include the following.

- Standard lead acid (also called flooded or wet cell)
 - Starting
 - Deep cycle
 - Serviceable or maintenance free
- Maintenance free absorbent glass mat
- Maintenance free gel cell

Battery supplier service literature will provide specific removal and installation, maintenance, inspection, and testing requirements. See Original Equipment Manufacturer (OEM) service information

2.4 Battery Cables and Connections - Remove and Install

Tools Summary		
Part Name	Part Number	Quantity
Battery Brush		1

Tools Summary		
Part Name	Part Number	Quantity
Wire Brush		1
Clean Rag		1
Battery Terminal Sealant		1

General Information

There are different types of battery connections.

These instructions are representing basic safety practices when performing battery servicing.

Refer to OEM Service Manual for specific battery instructions

Remove

Make sure the battery compartment is ventilated.

Remove the cables in the following order.

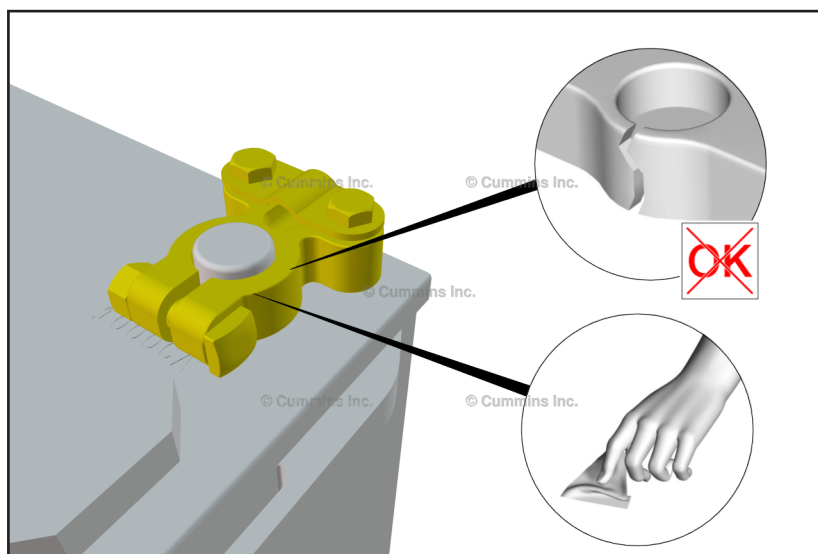
1. Negative Cable
2. Positive Cable

Remove OEM components, as necessary.

Clean

	WARNING	Battery Exposure Battery may explode. Battery acid may cause burns or injury. Avoid contact. Wear appropriate personal protective equipment.
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Clean the battery cables and connections if corroded.



wg542eyn

Use the following tool as applicable.

Tool Name	Tool Type	Part Number	Quantity
Battery Brush	Generic		1
Wire Brush	Generic		1
Clean Rag	Generic		1

Clean cable connections until surfaces are clear of debris.

Clean battery terminals until surfaces are clear of debris.

Clean battery tops until surfaces are clear of debris.

Check

Check the following.

- Battery cables and connections
- Terminals
- Connectors
- Cables

Replace if any of the following are found.

- Broken
- Cracked
- Deformed Terminals
- Loose Terminals

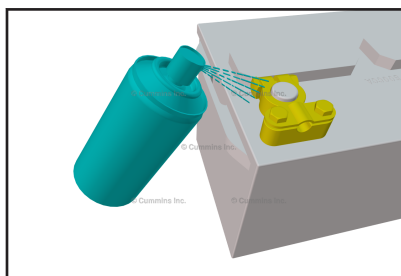
Install

Install the cables in the following order:

1. Positive Cable
2. Negative Cable

Tighten the battery connections.

Prevent corrosion by applying sealant to the terminals.



rag2bwbg

Tool Name	Tool Type	Part Number	Quantity
Battery Terminal Sealant	Generic		1

Tighten the battery connections.

- Refer to OEM manual for tightening steps.

2.5 Equipment - Safe Service

General Information

Lock out / Tag out must only be performed by persons authorized to perform Lock out / Tag out on the specific equipment.

Lock-Out-Tag-Out

Comply with location specific lock out / tag out procedures including, but not limited to the following.

- Permits for work
- Emergency stop process
- Immobilization of vehicles or equipment
- Lock out / tag out signage and required data
- Informing appropriate site personnel
- Application back-up dependencies
 - Example: Critical life support generator sets

Shut off all sources of energy and check that all moving parts have stopped moving.

Use appropriate personal protective equipment.

Electrical equipment must be treated as energized until proven de-energized with a live-dead-live test

Activate emergency stop.

Isolate the equipment from all energy sources, but not limited to the following.

- Generator set
- Control panels
- Associated equipment
- Batteries
- AC power to all associated heaters

Lock out and apply tag to all points of isolation.

Lock out tags must include the following information at a minimum.

- Identity of the equipment being isolated
- Reason for isolation
- Date of the isolation
- Person that applied the lock out
- Method to contact the person

Verify the Lock out / Tag out by attempting to restart the equipment using manual controls.

Maintain the lock out by performing the following.

- Keep keys secure

Keep unqualified personnel away from the point of work

Re-Energize

Make sure that the work is complete.

Check the following.

- All connections have been restored

- All guards and covers are in place

Make sure that the area is clear of personnel and tools.

Close any test points or stop cocks that were opened during verification.

Cancel any work permits.

Notify the appropriate personnel that the lock out will be removed.

Remove the locks and tags.

Make sure the equipment is ready.

Chapter 3 - Basic Service Practices

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3.1 General Cleaning - Generator Set - Overview

General Information

This document provides general cleaning practices for a generator set.

Detailed information for individual components can be found in the clean action for each component.

The intent of cleaning is to remove debris; this does not mean they have to appear as new.

Debris buildup can affect the performance and heat dissipation of pumps and other electric motors.

The general working area must be cleaned prior to opening closed systems.

Accumulation of oils and coolants can result in unsafe working conditions.

Overview

The following conditions must be met before cleaning is started.

- Sources of power must be isolated
- The generator set is not running.
- The generator set has cooled to ambient temperature
- Make sure drainage flows away from the generator set
- Ambient conditions are above freezing temperature

Low pressure wash should be used to remove dust and light debris.

Low pressure wash is defined as follows.

- Use water
- Flow rate between 0.07 and 0.15 lit/sec [1 and 2 gal/min]
- Diameter of the water outlet less than 15.9 mm [5/8 inch]
- Water temperature between 10 °C and 30 °C [50 °F and 86 °F]
- Spray from a distance greater than 10 cm [4 in]
- Spray the target component with a sweeping motion

Do NOT low pressure wash the following components.

- Air filters
- Air inlets
- Fans
- Belts
- Bearings
- Batteries
- Any exposed electrical terminals

High pressure wash should be used to remove large debris or other buildup not removed by a low-pressure wash.

High pressure wash is defined as follows.

- Use water
- Water pressure greater than 10340 kPa [1500 psi]
- Spray from a distance greater than 15 cm [6 in]
- Spray the target component with a sweeping motion

Do NOT high pressure wash the following components.

- Air filters
- Air inlets
- Fans
- Belts and Hoses
- Bearings
- Batteries
- Electrical Components
 - Wiring harnesses
 - Electronic Control Module
 - Electronic Control Module Connectors
 - Capacitive Coil Driver Module
 - Ignition Coils and Leads
 - Sensors
 - Any electronic actuators
- Electrical boxes

	WARNING	Steam Cleaning Risk of burns and flying debris. Wear appropriate personal protective equipment.
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Steam cleaning can be used to remove debris or buildup not removed by a low-pressure wash.

Steam cleaning is defined as follows.

- Use water
- Temperature between 99 °C and 110 °C [210 °F and 230 °F]
- Spray the target component with a sweeping motion

Do not steam clean the following components:

- Air filters
- Air inlets
- Fans
- Belts and Hoses
- Bearings
- Batteries
- Electrical Components
 - Wiring harnesses
 - Electronic Control Module
 - Electronic Control Module Connectors
 - Capacitive Coil Driver Module
 - Ignition Coils and Leads
 - Sensors
 - Any electronic actuators
- Electrical boxes

	CAUTION	Solvent Exposure Skin and eye irritation may occur. Follow manufacturer's instructions for use. Wear appropriate personal protective equipment.
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Solvent and acid cleaning should be used to remove oils or other buildup not removed by a low-pressure wash.

Solvent cleaning should only be used on focused areas.

There are several solvent and acid-type cleaners available.

Follow the manufacturer's recommendations for cleaning solvents.

Do NOT use solvents on the following components.

- Rubber couplings
- Rubber vibration isolators

Chapter 4 - Service Support

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4.1 Additional Service Literature - Overview

Overview

The following publications and information is available online for this Package.

- HGLAA Installation manual (A062Y945)
- HGLAA Operator manual (A062Y985)
- HGLAA Parts manual (A062Y987)
- HGLAA Recommended Spares List (A062Z023)
- HGLAA Service Suite
 - HGLAA Service manual (A062Y986)
 - HGLAA Troubleshooting and Diagnostics manual (A065E615)
- Quick Start Guide (RV/CM) (A062P264)
- Health and Safety manual (0908-0110)
- Service Tools manual (A043D529)
- Warranty manual (4021290)
- PGBU Global Product Coverage manual (A048H943)

The following publications are available to be purchased in hard copy for this Package.

- HGLAA Installation manual (A062Y945)
- HGLAA Operator manual (A062Y985)
- HGLAA Service manual (A062Y986)
- HGLAA Parts manual (A062Y987)
- HGLAA Literature List (A062Y988)
- HGLAA Standard Repair Times manual (A062Z021)
- HGLAA Recommended Spares List (A062Z023)
- Quick Start Guide (RV/CM) (A062P264)
- RV Generator Set Warranty Statement (OOAB-2012-03)
- Health and Safety manual (0908-0110)
- Service Tools manual (A043D529)
- Warranty manual (4021290)
- PGBU Global Product Coverage manual (A048H943)

4.2 Service Assistance - Overview

Overview

To contact or locate the nearest Cummins® Authorized Repair Location, please visit locator.cummins.com.

- For 24 hour assistance, please visit www.cummins.com/support

When contacting Cummins Inc., have the following information readily available:

- Model
- Specification
- Serial Number

Cummins Inc. provides a 24-hour, toll free telephone number to aid in technical and emergency service when a Cummins® Authorized Repair Location can not be reached or is unable to resolve an issue with a Cummins® product.

U.S. and Canada

- 1-800-CUMMINS™ (1-800-286-6467)

Mexico

- 01-800-CUMMINS (2866467)
- Email: contacto.mexico@cummins.com

Australia/New Zealand

- For Australia: 1-300-CUMMINS (1300 286 6467)
- For New Zealand: 0-800-CUMMINS (0800 286 6467)
- Email: 1300cummins@cummins.com

Brazil

- 0800 CUMMINS (0800 286 6467)
- Email: falecom@cummins.com

Europe

- From the UK, Ireland, Germany, France, and the Netherlands - Freephone: 00 8000 2866467 or +44 1327 886464
- Email: cceurope@cummins.com

China

- (+86) 400 810 5252
- Email: cac@cummins.com

India

- 800 210 2525
- Email: powermaster-India@cummins.com

Russia

- Primary: +7 495 926 8624
- Alternate: +7 962 943 1211

For All Other Areas:

- +44 (0) 1327 886464

4.3 Manufacturing Locations - Overview

Overview

U.S. and Canada

- Cummins Inc.
- 1400 73rd Ave. NE
- Minneapolis, MN 55432 USA
- 1-800-CUMMINS™ (1-800-286-6467)
- Phone +1 763-574-5000
- Fax +1 763-574-5298

EMEA, CIS

- Cummins Inc.
- Royal Oak Way South
- Daventry
- Northamptonshire
- NN11 8NU
- United Kingdom
- Phone +44 1327 88-6453
- Fax +44 1327 88-6125

Brazil

- Cummins Inc.
- Rua Jati, 310, Cumbica
- Guarulhos, SP 07180-900
- CNPJ: 43.2201.151/0001-10
- Brazil
- Phone 0800 286 6467

China

- Cummins Inc.
- No.118 South Quanli Road
- Wuhan Economic & Technological Development Zone
- Hubei, P.R.China 430058
- Phone + 86 (27) 8421 4008
- Fax + 86 (27) 8421 4804

India

- Cummins Inc.
- Plot No B-2, SEZ Industrial Area
- Village-Nandal & Surwadi, Taluka- Phaltan
- Dist- Satara, Maharashtra 415523
- India
- Phone +91 021 66305514

LATIN AMERICA

- Cummins Inc.
- 3350 Southwest 148th Ave.
- Suite 205
- Miramar, FL 33027
- USA
- Phone +1 954 431 551
- Fax +1 954 433 5797

Mexico

- Cummins Inc.
- Eje 122 No. 200 Zona Industrial
- San Luis Potosi, S.L.P. 78395
- Mexico

- Phone +52 444 870 6700
- Fax +52 444 824 0082

ASIA PACIFIC

- Cummins Power Generation (S) Pte Ltd.
- 85 Tuas South Avenue 1
- Singapore 637419
- Fax +65 6265 6909

Chapter 5 - Product and System Identification

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5.1 Model Identification - Generator Set - Overview

General Information

Model names provide identification information for Cummins® products.

Information that may be found in model names are listed below.

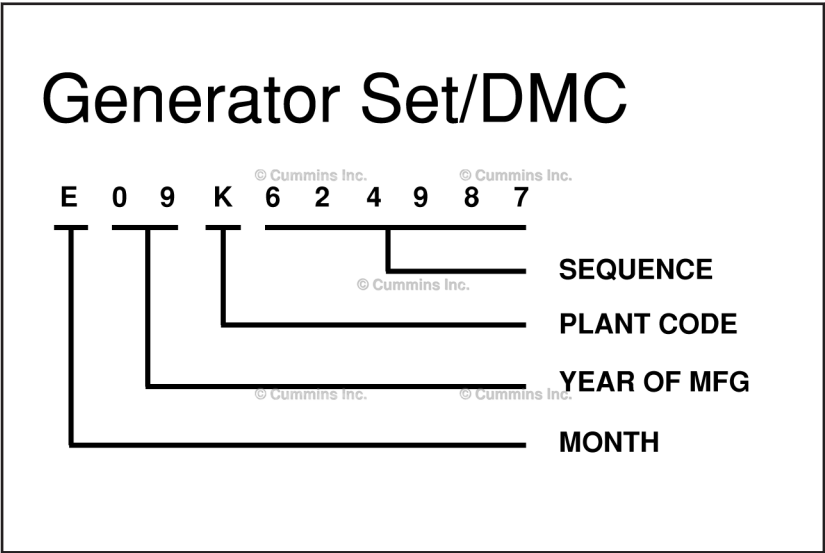
- Marketing Model Names and numbers
- Specification numbers
- Electrical characteristics
 - Control system

When contacting Cummins® dealers or distributors for service assistance, always be ready to provide the complete product model, service model name or serial number information for the product as listed on the nameplate.

Overview

Generator set model names consist of the following.

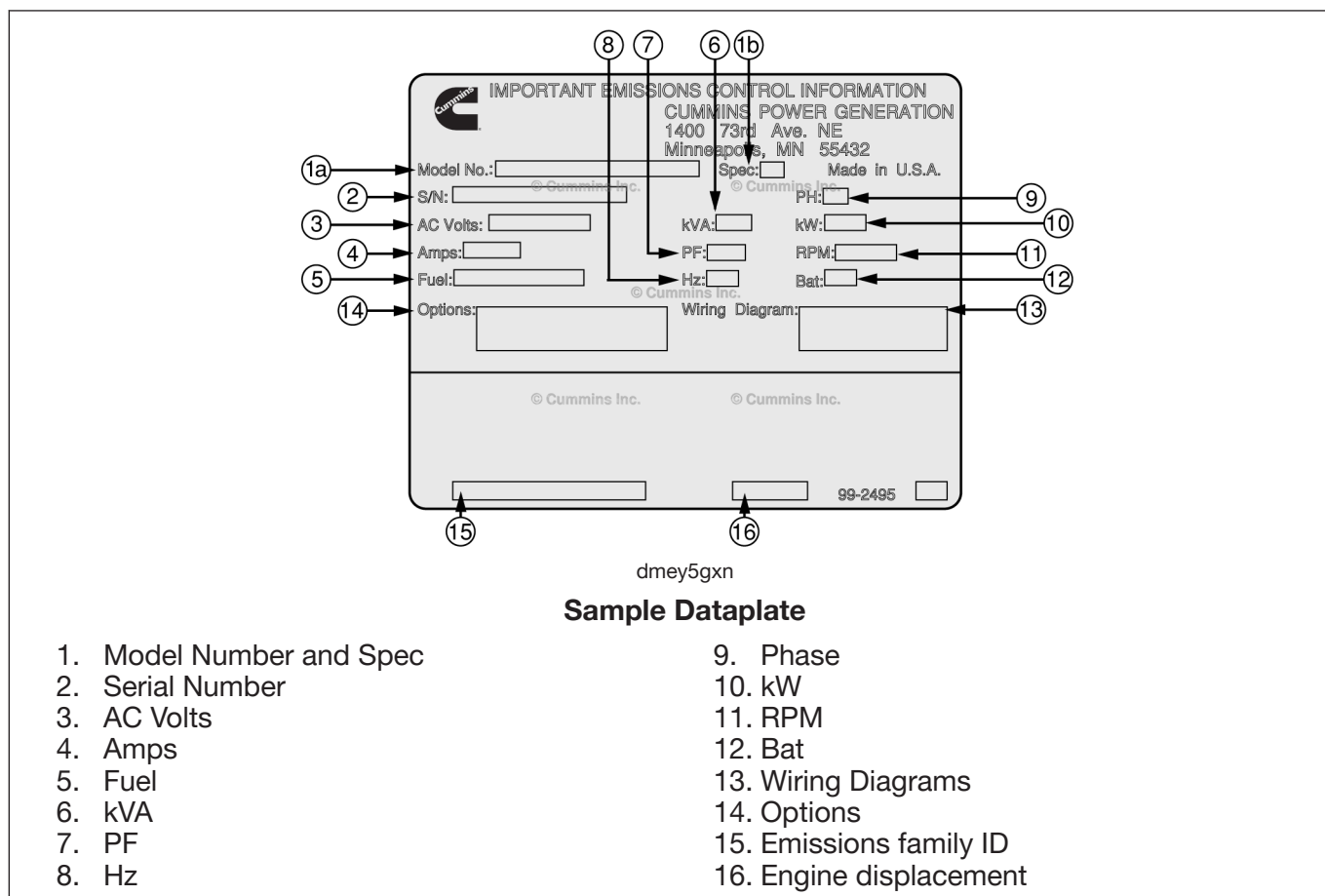
1. Month
2. Year of manufacture
3. Plant code
 - 0 = Fridley USA
 - A = Australia
 - B = Italy Rental
 - C = Italy (DIEM)
 - D = Daventry UK
 - E = Loncin
 - F = Domestic US
 - G = Girtz & Brazil G-Drive
 - H = Wuhan China
 - I = India
 - J = Cal Pac (Eco systems)
 - K = Kent UK
 - M = SLP Mexico
 - Q = CPT China
 - S = Singapore
 - T = Brazil
 - V = Venture
 - W = Wuxi China
 - WH = China Domestic Products
 - Y = Nigeria
 - Z = GE Zenith Controls
4. Sequence - 6 characters
 - Product since 2012 have 6 numeric characters previous to this it was three letters followed by three numbers



nnbxp4xn

5.2 Dataplates - Overview

Diagrams



General Information

Each generator set is provided with a dataplate that contains the generator set model and serial number.

This information is needed when contacting Cummins for parts, service, and product information.

The last character of the Model Number is the "Spec Letter" and is important for obtaining the correct information.

Overview

The fields on the dataplate refer to the following information.

- Model Number and Spec Letter
- S/N
 - Serial Number
- AC Volts
 - Generator set output voltage rating
- Amps
 - Generator set output amps at rated kW and maximum volts